

Lab Task#6

Sec A

Note: All statements should be commented. Plagiarism will get straight zero.

Use lists, tuples, sets and dictionaries. Do not import any module.

Question 1

Quiz results of a class are stored as a list of tuples. Each tuple is (roll_no, marks) and the quiz is out of 20:

```
results = [("24P-0101", 14), ("24P-0102", 9), ("24P-0103", 18),  
("24P-0104", 11), ("24P-0105", 18), ("24P-0106", 6), ("24P-0107", 15)]
```

Do not change the given list. Loop over the tuples to:

- 1) Compute and print the class average.
- 2) Find the highest marks and print every roll number that scored it (more than one student may have the highest marks).
- 3) Build a list below_average containing the roll numbers of students who scored below the average. Print the list and how many students are in it.
- 4) Build a list of only the marks, sort it from highest to lowest, and print the top 3 marks using slicing.

Question 2

Students register for the Python and ML courses through an online form. Some students submitted the form more than once:

```
python_forms = ["24P-0101", "24P-0102", "24P-0103", "24P-0102",  
"24P-0104", "24P-0101", "24P-0105"]
```

```
ml_forms = ["24P-0103", "24P-0106", "24P-0101", "24P-0107", "24P-0106",  
"24P-0105"]
```

Choose a suitable data structure so that each student is counted only once, then:

- 1) For each course, print the number of forms submitted and the number of unique students.
- 2) Print the students registered in both courses.
- 3) Print the students registered in Python but not in ML.
- 4) Print how many different students are registered in total (in either course).
- 5) Student 24P-0105 drops ML. Update your data, print the ML students again, and print whether 24P-0105 is still registered in any course.

Print every group of students as a sorted list.

Question 3

Votes for class representative are stored in a list:

```
votes1 = ["Ali", "Sara", "Ali", "Hamza", "Sara", "Ali", "Zara",  
"Hamza", "Sara", "Sara"]
```

Write `count_votes(votes)` that returns a dictionary mapping each candidate to their number of votes. Build the dictionary with a loop (do not use the `count()` method).

Write `announce_result(votes)` that uses `count_votes(...)` and prints:

- 1) each candidate with their votes
- 2) the winner, or "Tie between:" followed by the names if two or more candidates share the highest votes
- 3) the candidates who received fewer than 2 votes

Include a docstring in each function. Call `announce_result` for `votes1`, and again for:

```
votes2 = ["Ali", "Sara", "Sara", "Ali", "Zara"]
```

Question 4

A stationery shop keeps its price list (in rupees) in a dictionary:

```
prices = {"pen": 30, "notebook": 120, "bag": 1500, "calculator": 950,
"marker": 60}
```

A customer's cart is a list of tuples (item, quantity). The same item may appear more than once, and some items are not sold by the shop:

```
cart = [("pen", 3), ("notebook", 2), ("marker", 1), ("pen", 2),
("eraser", 4), ("bag", 1), ("glue", 2)]
```

- 1) Combine the cart into a dictionary quantities that maps each item to its total quantity. Print it.
- 2) Collect the items that are not in the price list in a set named unavailable and print them as a sorted list. These items are not billed.
- 3) For every available item print: item x quantity = line total, and calculate the total bill.
- 4) If the total is more than 1800 rupees, apply a 10% discount. Print the total, the discount and the amount to pay.
- 5) Print the item with the highest line total and its amount.